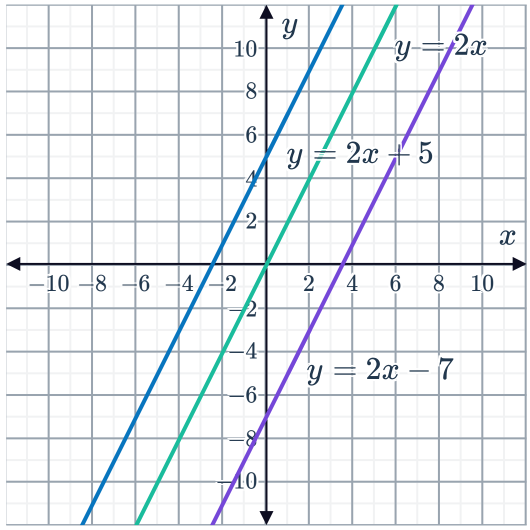
# Parallel lines

1 The equations $y = 2x$, $y = 2x + 5$ and $y = 2x - 7$ have been graphed on the same number plane:




a What do all of the equations have in common?

b What do all the lines have in common?

2 State whether the following pairs of lines are parallel:

a $y = -3x - 2$ and $y = -3x + 9$

b $y = -2x - 5$ and $y = -2x - 8$

c $y = 7x + 8$ and $y = -5x + 8$

d $y = 4x - 1$ and $y = 4x - 6$

e $x = 4$ and $y = 5$

f $y = 7x - 5$ and $y = -7x + 6$

g $y = 3x - 5$ and $y = -\frac{1}{3}x + 6$

h $y = -8x - 2$ and $9x + 7$

3 State whether the following lines are parallel to $y = 7x + 3$:

a $y = 7x - 3$

b $y = 6x + 3$

c $y = 7x + 4$

d $y = 7x$

e $y = 6x + 4$

f $y = -7x + 3$

g $y = 3x + 7$

h $y = -3x - 7$

4 State whether the following lines are parallel to $y = -3x + 2$:

a $y = 3x$

b $-3y - x = 5$

c $y = -10 - 3x$

d $y + 3x = 7$

5 State whether the following lines are parallel to $y = 9x + 2$:

a $y = 9x$

b $y = -9x + 5$

c $y = -9x + 2$

d $y = 9x - 2$

6 Consider the lines $L_1 : 5x + 8y - 10 = 0$ and $L_2 : y = -\frac{5x}{8} + 8$.

a Find the gradient of line $L_1$.

b Find the gradient of line $L_2$.

c Are the two lines parallel?

7 Consider the line $y = 2x + 2$. If every point on the line is shifted $2$ units up, find the equation of the new line.

8 Every point on a particular line is shifted $6$ units down. The resulting line has equation $y = 2x - 2$. Find the equation of the original line.

9 Find the equation of a line described by the following information:

a Parallel to the $x$-axis and passes through $(-10, 2)$.

b Parallel to the $y$-axis and passes through $(-7, 2)$.

c Parallel to the line $y = -3x - 8$ and cuts the $y$-axis at $-4$.

d Parallel to the line $y = 8x - 3$ and cuts the $y$-axis at $5$.

e Parallel to the line $y = -2x + 9$ and passes through the point $(-3, 1)$.

10 If the following pairs of lines are parallel, find the value of $b$ :

a $0 = 5x - 9y + 6$
$0 = bx + 3y - 2$

b $y = bx + 4$
$2y = \dfrac{4}{5}x - 7$

11 The line $L_1$ goes through $(3, 2)$ and $(-2, 4)$.

a Find the gradient of line $L_1$.

b Find the equation of the line that has a $y$-intercept of $1$ and is parallel to line $L_1$.

12 The line $L_1$ goes through $(-9, 10)$ and $(2, -1)$.

a Find the gradient of line $L_1$.

b Find the equation of the line that passes through $(10, 8)$ and is parallel to line $L_1$.

13 The line $L_1$ passes through the point $(9, -5)$ and is parallel to the line $y = -5x + 2$.

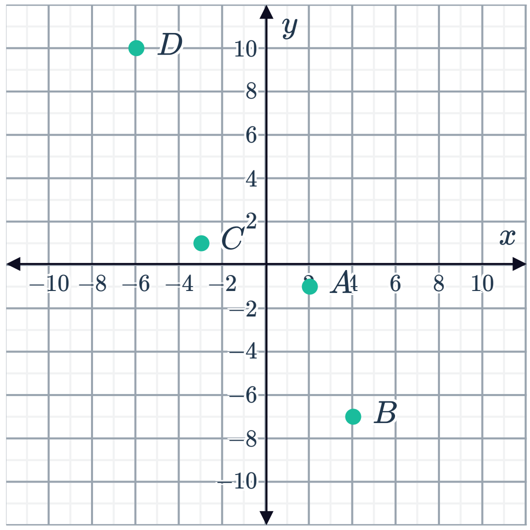
a Find the gradient of line $L_1$.

b Find the equation of line $L_1$.

## Applications

**14** Consider the following points on the number plane:

- Point $A\,(2,-1)$
- Point $B\,(4,-7)$
- Point $C\,(-3,1)$
- Point $D\,(-6,10)$



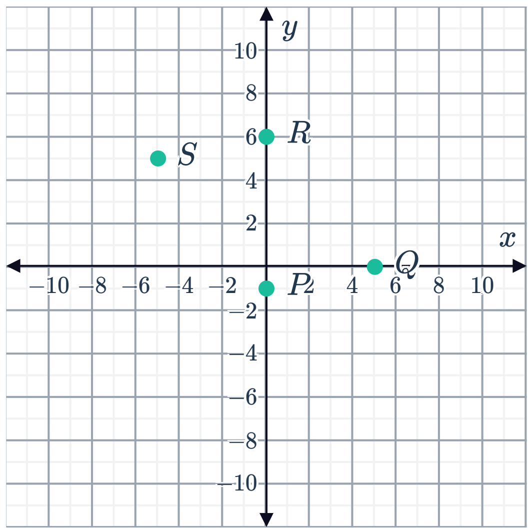

a Find the gradient of the line $AB$.

b Find the gradient of the line $CD$.

c Is the line $CD$ parallel to $AB$?

**15** Consider the following points on the number plane:

- Point $P\,(0,-1)$
- Point $Q\,(5,0)$
- Point $R\,(0,6)$
- Point $S\,(-5,5)$




a Find the gradient of $PQ$.

b Find the gradient of $RS$.

c Are $PQ$ and $RS$ parallel?

d Are $QR$ and $PS$ parallel?

e What type of quadrilateral is $PQRS$?

**16** Determine whether the following sets of points are collinear:

a Point $A\,(-4,3)$, Point $B\,(-2,7)$ and Point $C\,(-7,-3)$.

b Point $A\,(-3,1)$, Point $B\,(-2,6)$ and Point $C\,(0,14)$.

c Point $A\,(-2,4)$, Point $B\,(2,2)$ and Point $C\,(4,1)$.